

Lecture 4

Recap last time: Divisibility & Prime numbers.

For example: $(d, m : \mathbb{N})$ then $d \mid m$ (1dvd)

$$\Leftrightarrow \exists k, \text{ st. } m = dk.$$

Being prime is property of nat. numbers

in Lean: $(\text{Prime } p) : \text{Prop}$ (where $p : \mathbb{N}$).

Today Prime factorisation theorem.

Theorem let n be nat. number, $n > 0$.

There exist prime numbers $p_0, \dots, p_k$ ^{depend on n} and ^{depends on n}

natural numbers $e_0, \dots, e_k$ ^{depend on p_i 's.} st. $(e_i \neq 0)$.

$$n = \prod_{i=0}^k p_i^{e_i} = p_0^{e_0} \dots p_k^{e_k}$$

Moreover, these prime numbers & exponents are unique.

Idea: How do formalize the statement efficiently?

We encode the factors and their exponents simultaneously using a function

$$\text{PExp}(n) : \{\text{Primes}\} \longrightarrow \mathbb{N}$$
$$p \longmapsto \max \{e \mid p^e \mid n\}.$$

Prop: If $p \nmid n$ then $\text{PExp}(n)(p) = 0$.

- There are only fin. many primes st. $\text{PExp}(n)(p) \neq 0$.

(Because $p \mid n \Rightarrow p \leq n$ there are only fin. many primes dividing n).

Simplification: We extend the above function to all nat. numbers as follows

$$\text{PExp}(n) : \mathbb{N} \longrightarrow \mathbb{N}$$
$$p \longmapsto \begin{cases} \max \{e \mid p^e \mid n\} & \text{if } p \text{ is prime} \\ 0 & \text{otherwise.} \end{cases}$$

Restate theorem: let $n \in \mathbb{N}$, $n > 0$.

1) We can write $n = \prod_{\substack{p \in \mathbb{N}, \\ \text{PEXP}(n)(p) \neq 0}} p^{\text{PEXP}(n)(p)} \longleftrightarrow (\text{PEXP}(n)).\text{support}$

2) The map $\{n \in \mathbb{N} \mid n > 0\} = \mathbb{N}_+ \longrightarrow \text{Fun}(\mathbb{N} \rightarrow \mathbb{N})$
 $n \longmapsto \text{PEXP}(n)$

is injective.

Ex. $n = 60$.

$$2 \mid n$$

$$2 \cdot 30 = 60$$

$$3 \mid n$$

$$3 \cdot 20 = 60$$

$$5 \mid n$$

$$5 \cdot 12 = 60$$

1) $\text{PEXP}(60)(p) = 0$ if $p \neq 2, 3, 5$.

2) Find actual values: $2 \mid 60$, $4 \mid 60$

$$4 \cdot 15 = 60$$

$\text{PEXP}(60)(2) = 2$ but $8 \nmid 60$.

$$\text{PEXP}(60)(3) = 1$$

$$PExp(60)(5) = 1.$$

$$3) \quad 60 = 2^2 \cdot 3^1 \cdot 5^1 = 4 \cdot 15 = 60.$$

$$PExp : \mathbb{N} \rightarrow \text{Fun}(\mathbb{N}, \mathbb{N}).$$

For every number n , we get a function computing its prime exponents.

Proof of Uniq. assuming 4). (Decomp. part of The theorem)

Let $n, m \in \mathbb{N}$, $m \neq 0 \neq n$.

Assume $PExp(n) = PExp(m)$.

Want to show: $n = m$.

Step 1) $PExp(n) = PExp(m)$ we also have

$$\{ p \mid \underline{PExp(n)}(p) \neq 0 \} = \{ p \mid \underline{PExp(n)}(p) \neq 0 \}.$$

Step 2) $\forall p \in \{ p \mid \underline{PExp(n)}(p) \neq 0 \}.$

$$PExp(n)(p) = \underline{PExp(n)}(p).$$

$$n = \prod_{p \in \{ p \mid \underline{PExp(n)}(p) \neq 0 \}} p^{PExp(n)(p)} = \prod_{p \in \{ p \mid PExp(n)(p) \neq 0 \}} p^{PExp(n)(p)}$$

Mul. is commutative.

Notation: $(PExp(n))_{sup} = \{ p \mid PExp(n)(p) \neq 0 \}.$

Proof of part 1) of theorem: Strong induction on n .

Base case: $n = 0$ trivial.

$$\underline{n \geq 0}$$

Induction hypothesis: theorem holds $\forall m \leq n$.

We want to show that theorem holds $(n+1)$.

Case $n=0$: $(n+1)=1$ in the case theorem

$$\left[1 = \prod_{\{p \mid \text{PExp}(1)(p) \neq 0\}} \text{PExp}(1)(p) = 1. \right]^{1)}$$

$\xrightarrow{\text{By defn.}}$

using $\text{PExp}(1) = \text{constant } 0$.

Case $n > 0$:

Observation: $(n+1) > 1$, then there exists a

prime number $\boxed{q \mid (n+1)}$ $\rightarrow$ library (Exc. Strong induction)

in other words $\boxed{q \in (\text{PExp}(n) / \text{supp})}$ $\rightarrow$ technical step. (2)

Then we can show:

$$(n+1) = q^{\text{PExp}(n+1)(q)} \cdot k$$

for $k \in \mathbb{N}$

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$$k \leq n$$

We need to know $k < n+1$ (Exercise 0)

by induction hypothesis we know

$$k = \prod_{p \in (\text{PExp}(k).supp)} p^{\text{PExp}(k)(p)}$$

Summary:

$$(n+1) = q^{\text{PExp}(n+1)(q)} \cdot \prod_{p \in (\text{PExp}(k).supp)} p^{\text{PExp}(k)(p)}$$

(*)

What do we need to prove:

[1) $(\text{PExp}(n+1)).supp = (\text{PExp}(k)).supp \cup \{q\}$ ^{disjoint}

[2) $\text{PExp}(n+1)(p) = \text{PExp}(k)(p)$ if $p \neq q$.

Then we can write ~~*~~ as

$$\begin{aligned} (2) \quad & \prod_{p \in \text{PExp}(n). \text{Supp}} \text{PExp}(n+1)(p) \stackrel{(1)}{=} \prod_{p \in (\text{PExp}(n+1)). \text{Supp}} \text{PExp}(n+1)(p) \end{aligned}$$

Lemma 1) $(\text{PExp}(1)). \text{Supp} = \emptyset$.

Pf. 1 has no prime factors.

Lemma 2) $(n > 1) \quad (\text{PExp}(n)). \text{Supp} \neq \emptyset$

Pf. Every $n > 1$ has at least one prime factor.

Lemma 3) $(n > 1) \quad q \in (\text{PExp}(n)). \text{Supp}$

$$n = q^{\text{PExp}(n)(q)} \cdot k$$

$\Leftrightarrow (n \mid q^e / n^3), \quad \text{for some } k.$

Lemma 4 Same assumptions

$$(\text{PExp}(n)).\text{support} = (\text{PExp}(k)).\text{support} \\ \perp \{d, q\}.$$

Lemma 5) Same assumptions.

If $p \neq q$. Then

$$\text{PExp}(n)(p) = \text{PExp}(k)(p).$$

Lemma 4 1) q^e and k are
coprime (for $e = \text{PEXP}(n)(q)$.)

2) If $p \mid n \iff$

$p = q$ or $p \mid k$.

[If p is prime number
then $p \mid n \cdot m \iff p \mid n \vee p \mid m$]

Lemma 5 If $p \mid k$ then

$$\max \{e | p^e | k\}$$

$$= \max \{e | p^e | q^{\text{Exp}(u)(q)} \cdot k\}.$$
